

Auditing and Repairing Multiscale Coordinate Projection in Frozen Visual Time-Series Anomaly Detection

Anonymous Authors

Abstract—Frozen visual foundation models offer a training-free screen for telemetry anomalies, but multiscale token evidence must be returned to the base grid that generated it. We audit the released ViT4TS projector and find that, at each pooled scale, all 195 valid support queries are displaced by one flattened cell: 13 wrap across rows, while the terminal query is unsupported. We implement Index-Consistent Harmonic Projection (IHP), a minimal coordinate-contract repair that interprets supplied memberships literally before applying the unchanged ViT4TS harmonic projector. The renderer, frozen encoder, all-window visual memory, matching, scale fusion, and temporal stitcher remain fixed. On 492 series from eleven NAB, NASA, and Yahoo subdatasets, IHP improves same-cache equal-subdataset F1-max, AUPRC, and VUS-PR by 0.0267, 0.0218, and 0.0102 over the released-coordinate control. Unadjusted hierarchical bootstrap intervals exclude zero for AUPRC and VUS-PR but not F1-max; they quantify sampling variability rather than post-selection inference. As secondary external context, IHP obtains F1-max 0.662 versus the paper-reported ViT4TS value 0.612 and is numerically higher on 9/11 subdatasets. The repair uses no training, labels, learned parameters, or additional encoder pass, showing that coordinate auditing can improve a frozen visual anomaly screen without increasing model capacity.

Index Terms—time-series anomaly detection, telemetry monitoring, visual foundation models, coordinate consistency, harmonic projection

I. Introduction

Training-free anomaly screening is attractive for heterogeneous telemetry, provided that detected evidence can be localized to the measurements that generated it. Cloud services, networked applications, spacecraft, and sensor systems produce signals with different normal regimes and failure patterns; maintaining a separately trained detector for every source is costly. Recent work therefore explores both reliable cross-domain benchmarks [1] and telemetry-specific unsupervised detectors [2]. Yet deployment requires more than a strong anomaly score: an operator must return that score to the correct timestamp for alert ranking and diagnosis.

Frozen vision representations provide a promising alternative to dataset-specific training. VLM4TS converts a time-series window to a line image, uses its ViT4TS stage to screen candidate anomalies with frozen CLIP features, and can then invoke a vision-language model for verification [3], [4]. The screening stage builds multiscale visual tokens, compares them with a visual memory, projects the resulting token costs to a common image grid,

and finally maps image columns back to time. This design makes the token-to-grid projector a semantic interface: even perfectly informative frozen tokens cannot localize an anomaly if their inverse coordinate map assigns evidence to a different cell.

We find that this coordinate contract is inconsistent in the released projector. The supplied pooling memberships contain zero-based flattened base-cell indices, but the projector queries whether each set contains $i + 1$. At each pooled scale on the 14×14 grid, 195 output queries find support, yet every one is displaced by one flattened cell. Thirteen of these queries wrap across row boundaries, and the terminal query has no support. A finite-value fallback then turns that structural hole into a zero anomaly contribution. The issue is therefore not the syntactic size of an index change: the shifted graph assigns multiscale evidence to incorrect cells and makes the projection incomplete for every rendered window.

We address this interface with Index-Consistent Harmonic Projection (IHP), a minimal coordinate-contract repair. IHP reconstructs the literal token-to-cell graph specified by the zero-based memberships and then applies the unchanged ViT4TS harmonic projector on that corrected support. The harmonic mean itself is not presented as new; the contribution is a complete projection defined on the coordinate convention that generated the masks. The renderer, encoder, all-window visual memory, token matching, scale fusion, and timestamp stitching remain frozen, so IHP adds neither learned capacity nor another backbone pass.

This narrow intervention enables two complementary evaluations. A label-free certificate tests whether the predicted displacement, row wraps, and hole exist and whether literal incidence restores complete coverage. A same-cache comparison then holds every visual token and downstream operation fixed while replacing only the projection. Across 492 series and all eleven NAB, NASA, and Yahoo subdatasets, IHP increases equal-subdataset F1-max, AUPRC, and VUS-PR over the released-coordinate control by 0.0267, 0.0218, and 0.0102. Unadjusted hierarchical intervals exclude zero for AUPRC and VUS-PR, whereas the F1-max interval crosses zero. As external context, IHP obtains 0.662 F1-max versus the paper-reported ViT4TS value of 0.612 and is higher on 9/11 subdatasets; this external comparison is descriptive, not

paired.

The contributions are:

- We formalize index consistency at the multiscale mask-to-grid interface and certify that all 195 valid pooled-scale queries are shifted, including 13 row wraps, while the terminal query is unsupported.
- We derive and implement a complete index-consistent repair for the ViT4TS mask-to-grid interface, without changing the inherited harmonic reducer or any learned component.
- We evaluate the intervention on 492 series from eleven telemetry subdatasets, separating an explicitly paper-reported external comparison from a same-cache causal control with hierarchical uncertainty estimates.

II. Related Work

A. Numerical and Pretrained Time-Series Anomaly Detection

Recent time-series anomaly detection (TSAD) methods improve the detector itself through patch representations, calibrated normal spaces, or richer temporal interactions. PaAno learns efficient patch-level representations, DoKnowAD refines normal representations with domain knowledge, and IGAD introduces a parameter-free idempotent reconstruction rule [5]–[7]. CrossAD and ScatterAD model cross-scale, cross-window, or temporal-topological structure [8], [9], whereas DADA and CATCH pursue generalization through pretrained bottlenecks or frequency-aware channel modeling [10], [11]. A recent analysis further shows that temporal understanding and precise localization remain difficult for large language models [12]. These advances change learned representations, reconstruction objectives, or detector semantics. IHP instead addresses the inverse interface after a frozen visual detector has already produced multiscale token costs.

B. Visual and Multimodal TSAD

Visual encodings provide a second route to transferable TSAD. Recent systems evaluate multimodal language models on rendered time series, learn visual conversions for anomaly recognition, or align visual evidence with temporal locations [13]–[15]. The closest system is VLM4TS, whose ViT4TS screening stage renders windows, extracts frozen visual features, matches them to a visual memory, and projects multiscale token costs to a dense field before reducing that field to a time series [3]. Its optional second stage uses a vision-language model to refine candidate intervals. Our evaluated scope is deliberately narrower: we retain the released renderer, frozen encoder, memory, matching, fusion, and stitching, and repair only the multiscale screen’s mask-to-grid projection. VETime is a concurrent preprint that inverts a periodic visual layout with learned alignment [15]; IHP instead uses the literal incidence already supplied by the frozen projector and adds no learned alignment.

C. Training-Free Localization and Grid Consistency

Training-free visual anomaly detectors demonstrate that strong frozen features can support dense localization through carefully chosen geometric operators. UniVAD matches component-aware patch features, LogSAD combines vision-language evidence, and SubspaceAD models a normal feature subspace without training a new backbone [16]–[18]. UniFormally additionally studies background masking and top-ratio aggregation for task-agnostic visual anomaly detection [19]. These methods operate on spatial images and do not define how a time-series renderer’s pooled memberships must be pulled back to the base token grid.

Grid geometry is not innocuous. Alias-Free ViT and shift-equivariant ViT identify tokenization, patch merging, attention, and positional encoding as sources of sensitivity to translations [20], [21]. They redesign the visual architecture to control input shifts; IHP leaves the backbone unchanged and enforces a different property—literal incidence between already-computed pooled tokens and output cells. Thus, the relevant gap is not another feature encoder or anomaly reducer, but an explicit and complete coordinate contract for the inverse multiscale projection.

D. Telemetry Monitoring and Evaluation

Reliable telemetry anomaly detection requires both heterogeneous benchmarks and deployment-aware evaluation. TSB-AD consolidates diverse anomaly families, while VUS supplies range-aware, threshold-independent metrics [1], [22]. Recent MSN work studies behavior-level network-service anomalies and unlabeled distribution drift [2], [23], underscoring the importance of stable screening across evolving sensing and networking data. These works motivate our telemetry setting but are not quantitative comparators. Taken together, the literature develops stronger representations, multimodal reasoning, and dense visual localization. Our narrower gap lies in the released ViT4TS mask-to-grid interface: IHP makes its supplied multiscale support graph explicit and restores complete base-grid support before the unchanged harmonic aggregation.

III. Index-Consistent Harmonic Projection

A. Overview

Given an univariate telemetry series, the frozen ViT4TS screen renders overlapping windows as line images, extracts multiscale visual tokens with a CLIP Vision Transformer (ViT), and compares the tokens with a median visual memory built from all windows of that series [3], [4], [24]. The resulting token costs must then be returned to a common spatial grid before they can be localized in time. Index-Consistent Harmonic Projection (IHP) changes only this return map. As shown in Fig. 1, IHP restores the literal token-to-cell support relation and evaluates the inherited harmonic projector unchanged on that corrected

support before the fixed scale fusion and timestamp stitcher.

The design follows a single causal chain. Section III-C defines the coordinate-contract failure. Sections III-D and III-E present the repair and its inherited reducer, and Section III-F specifies the frozen-system boundary and computational properties. IHP has no training objective, learned parameter, or label-dependent branch.

B. Frozen Screening Interface

Let a rendered window be encoded on a $g \times g$ base grid, where the frozen ViT-B/16 configuration gives $g = 14$ and $N = g^2 = 196$ base cells. We index a base cell by $i \in \Omega = \{0, \dots, N - 1\}$ and use $s \in \mathcal{S} = \{\text{b}, \text{m}, \text{l}\}$ for the base, medium, and large token scales. Their token counts are $K_{\text{b}} = 196$, $K_{\text{m}} = 169$, and $K_{\text{l}} = 144$, corresponding to the base grid and the released overlapping 2×2 and 3×3 pooling layouts.

For window w , let $d_{w,s,k} \geq 0$ denote the frozen cosine-dissimilarity cost of token k at scale s . IHP does not alter how this cost is obtained: the line renderer, CLIP weights, multiscale pooling, median memory, and unrestricted nearest-memory comparison are fixed. For each pooled token, the released assets also provide a set $M_{s,k} \subseteq \Omega$ containing the flattened base-cell indices that contributed to that token. These sets are the sole geometric input to IHP.

C. Failure Mechanism: A Shifted Support Graph

We call a multiscale projector index-consistent when the support used to score output cell i is exactly the set of tokens whose provided membership contains i . However, the released implementation queries the already zero-based memberships with a one-step shift. Its effective incidence matrix is

$$A_s^{\text{rel}}(i, k) = \mathbb{I}[i + 1 \in M_{s,k}], \quad i \in \Omega, \quad (1)$$

where $A_s^{\text{rel}}(i, k)$ indicates whether scale- s token k is used for output cell i , $M_{s,k}$ is the released zero-based membership set, and $\mathbb{I}[\cdot]$ is the indicator function.

Equation (1) advances every support query by one flattened cell. At each pooled scale, 195 of 196 queries return valid support, but all 195 are displaced. Thirteen shifts, at indices 13, 27, $\dots$, 181, wrap across a row boundary; the remaining valid queries move within a row. The terminal output cell $i = 195$ queries index 196, which cannot occur in $M_{s,k} \subseteq \{0, \dots, 195\}$ and therefore has no pooled-token support. A subsequent finite-value fallback converts this structural hole into a zero anomaly contribution. The mismatch is thus a localization failure at the frozen token-to-grid interface rather than a shortage of encoder capacity.

D. Literal Incidence Repair

IHP first lifts each provided membership set into a literal incidence matrix:

$$A_s(i, k) = \mathbb{I}[i \in M_{s,k}], \quad A_s \in \{0, 1\}^{N \times K_s}. \quad (2)$$

Here, $A_s(i, k)$ links base cell i to pooled token k , $N = 196$ is the number of output cells, and K_s is the number of tokens at scale s . The base scale uses the identity incidence $A_{\text{b}} = I_N$ because each base token already corresponds to one output cell.

Literal lifting is needed because a flattened index has meaning only under the coordinate convention that generated it. Equation (2) uses the stored convention directly and introduces neither a learned mapping nor a corrective offset. Consequently, it preserves every supplied token-cell relation while removing the one-cell displacement, including its 13 row wraps, in Eq. (1).

Complete-coverage property. Let $n_s(i) = \sum_{k=1}^{K_s} A_s(i, k)$ be the number of scale- s tokens incident to cell i . If the released pooling layout covers the base grid, $\bigcup_{k=1}^{K_s} M_{s,k} = \Omega$, then $n_s(i) \geq 1$ for every $i \in \Omega$. This follows directly because each covered index appears in at least one membership set. The supplied medium and large masks satisfy this premise, so literal lifting produces valid support for all 196 cells without interpolation or an artificial terminal value.

E. Inherited Harmonic Projection on Corrected Support

After support is restored, IHP pulls each pooled token cost back to every base cell that contributed to it. For a pooled scale $s \in \{\text{m}, \text{l}\}$, let $\mathcal{I}_s(i) = \{k : A_s(i, k) = 1\}$ denote the incident token set. The projected cost is

$$p_{w,s}(i) = \begin{cases} 0, & \min_{k \in \mathcal{I}_s(i)} d_{w,s,k} = 0, \\ \frac{n_s(i)}{\sum_{k \in \mathcal{I}_s(i)} d_{w,s,k}^{-1}}, & \text{otherwise.} \end{cases} \quad (3)$$

In Eq. (3), $p_{w,s}(i)$ is the scale- s cost at base cell i for window w , $d_{w,s,k}$ is the frozen nonnegative token cost, $\mathcal{I}_s(i)$ is the literal incident set, and $n_s(i)$ normalizes for the number of incident tokens. The first branch is the exact zero convention used by the implementation: an incident zero dissimilarity makes the harmonic cost zero, so no numerical ϵ or tunable floor is introduced.

The harmonic reducer and its support normalization are inherited from ViT4TS; IHP does not claim a new mean in isolation. Its evaluated output is different from the released path because the unchanged reducer is composed with the index-consistent support graph and is defined for every output cell. The normalization prevents cells covered by more overlapping pooled tokens from receiving a larger value merely because their support cardinality is larger.

The three frozen scales are fused with the released equal-scale average,

$$P_w(i) = \frac{1}{3} (d_{w,\text{b},i} + p_{w,\text{m}}(i) + p_{w,\text{l}}(i)), \quad (4)$$

where $P_w(i)$ is the final base-grid cost for window w , $d_{w,\text{b},i}$ is the base-token cost at cell i , and the remaining terms are the medium- and large-scale pullbacks from Eq. (3). Thus, no scale weight is learned or selected per dataset.

IHP: Coordinate-Contract Repair

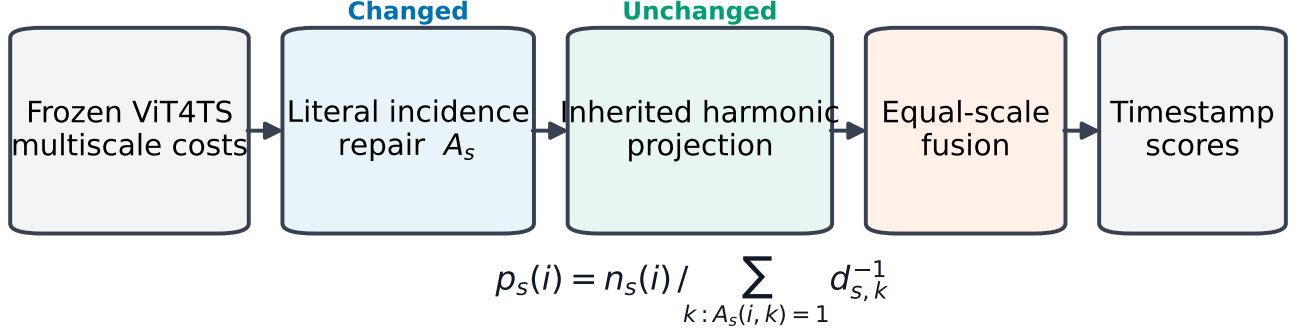


Fig. 1. IHP is a coordinate-contract repair inside the frozen ViT4TS screen. Literal incidence repairs the released zero-based support relation; the inherited ViT4TS harmonic projector then aggregates the same frozen token costs unchanged on that corrected support. The encoder, visual memory, matching, scale fusion, and timestamp stitching remain fixed.

F. Frozen-System Integration and Properties

The vector $P_w \in \mathbb{R}^{196}$ is reshaped to 14×14 , bilinearly upsampled to 224×224 , reduced with the released top-quarter column mean, and stitched across overlapping windows using the released temporal alignment. These operations, the anomaly threshold sweep, and the optional downstream VLM verification are outside IHP and remain unchanged. The present system evaluates the frozen visual screening stage; it does not attribute language-model reasoning to IHP.

IHP is deterministic, training-free, and parameter-free. Constructing the fixed incidence matrices and applying Eq. (3) has the same asymptotic projection complexity as the released membership queries. Most importantly, IHP reuses the same rendered windows, frozen visual tokens, memory, and token costs, so it adds no encoder forward pass. Test labels are read only by the evaluator after complete score files have been committed; they do not enter incidence construction, projection, fusion, or stitching.

IV. Experimental Protocol

A. Questions and Benchmark

The experiments answer three questions. First, when frozen visual evidence is held fixed, does index-consistent projection improve anomaly detection? Second, does label-free inspection confirm the predicted support-graph failure? Third, how does the repaired screen compare descriptively with the closest 2026 visual screen under its published benchmark taxonomy?

We evaluate all 492 univariate series used by the eleven subdatasets in VLM4TS Table 1 [3]. The suite contains five NAB groups (Artificial, AWS, AdExchange, Traffic, and Tweets), two NASA spacecraft groups (MSL and SMAP), and four Yahoo S5 groups (A1–A4). Their respective series counts are 6, 17, 5, 7, 10, 27, 53, 67, 100, 100, and 100.

These sources span synthetic events, cloud and application telemetry, spacecraft sensor channels, and Web-service time series. Each metric is first averaged over series within a subdataset and then equally over the eleven subdatasets, preventing the four 100-series Yahoo groups from dominating the aggregate.

B. Comparison Boundary and Selection Provenance

The causal comparison uses two score arms produced from identical frozen token caches. REL-U applies the released shifted-membership projection. IHP is the pre-specified IDX-U component arm and replaces only that projection with Eqs. (2) and (3). Both arms share rendered windows, CLIP tokens, multiscale pooling, visual memory, token matching, scale fusion, timestamp stitching, and the evaluator. F1-max selects its maximizing operating point separately for each arm; AUPRC and VUS-PR are threshold-free. Their paired difference therefore isolates score changes caused by the token-to-grid projection, not a representation or capacity change.

IHP was selected after the registered composite Coordinate Envelope failed its gate on the same 492 series. It was not created or retuned after that failure, but this promotion is still a post-selection step. Consequently, the hierarchical bootstrap intervals below quantify sampling variability and are not adjusted for arm selection; we do not interpret them as confirmatory post-selection inference.

The external reference is the ViT4TS screening row reported in VLM4TS Table 1. We transcribe its eleven F1-max values and reported equal-subdataset mean; we neither call them a local reproduction nor use them for paired inference. Full VLM4TS reports 0.659 F1-max, but it includes a separate language-model verification stage and is therefore only external system context for the locally evaluated screen.

C. Metrics and Uncertainty

We report F1-max for compatibility with VLM4TS Table 1. The primary same-cache evidence uses area under the precision-recall curve (AUPRC) and the range-aware volume under the precision-recall surface (VUS-PR) [22], which do not depend on a selected anomaly threshold. Higher values are better for all three metrics.

We use 10,000 hierarchical paired bootstrap replicates. Each replicate first samples the eleven subdatasets with replacement and then samples paired series within every selected subdataset. Percentile 95% intervals are reported for the IHP-minus-REL-U difference with seed 2027. As noted above, these intervals are unadjusted for the component-arm selection step.

D. Implementation, Offline Protocol, and Label Isolation

The frozen vendor repository is fixed at commit 8ab8c16414eb2c1a861dfc3e76f458180035a879. The screen uses CLIP ViT-B/16 with 224×224 images and a 14×14 base-token grid. Windows have length 240 and step 60. The released preprocessing detrends each complete series and applies full-series min-max scaling; visual memory is the token-wise median over all windows of the evaluated series. The score evaluator retains the released exponentially weighted moving average span $\max(1, \lfloor 0.01T \rfloor)$ and alpha grid $\{0.1, 0.01, 0.001\}$, and VUS uses maximum range window 100. Yahoo S5 is drawn from the verified mirror revision ea89753da2b451928436adb333c7a2e892461c7d.

Embedding extraction ran with Python 3.10, PyTorch 2.9.1+cu128, and one NVIDIA RTX 4090 with 24 GB of memory. IHP itself is deterministic and has no hyperparameter, training epoch, or dataset-specific choice. It operates on already computed token costs and requires zero additional encoder passes. Because preprocessing and memory use the complete evaluated series, the protocol is label-free but offline and transductive, not streaming. Labels are loaded only after complete score files have been written; neither cache construction nor either projection arm can access them.

V. Results and Analysis

A. Same-Cache Projection Ablation

Table I is the primary causal comparison. Relative to REL-U, IHP raises equal-subdataset F1-max from 0.6354 to 0.6621, AUPRC from 0.2974 to 0.3192, and VUS-PR from 0.6873 to 0.6975. The observed differences are +0.0267, +0.0218, and +0.0102. The unadjusted intervals exclude zero for AUPRC and VUS-PR, while the F1-max interval extends to -0.0023. These intervals describe sampling variability under the paired benchmark hierarchy and are not adjusted for the component-arm selection disclosed in Section IV.

The subdataset distribution shows that the ranking gains are not confined to one family. IHP improves AUPRC on 9/11 groups, with its largest gains on

TABLE I

Same-cache projection ablation on 492 paired series. Intervals use 10,000 hierarchical bootstrap replicates and are unadjusted for arm selection. “Positive” counts subdatasets with a positive IHP-minus-REL-U difference.

Metric	REL-U	IHP	Δ	95% interval	Positive
F1-max	.6354	.6621	+0.0267	[-.0023, .0681]	6/11
AUPRC	.2974	.3192	+0.0218	[-.0062, .0398]	9/11
VUS-PR	.6873	.6975	+0.0102	[-.0021, .0197]	7/11

TABLE II

Complete F1-max context for the VLM4TS Table 1 taxonomy. ViT4TS^P is external and paper-reported; REL-U, IHP, and Δ_{pair} are local same-cache results. Cross-environment columns are not a paired ranking.

Subdataset	n	ViT4TS ^P	REL-U	IHP	Δ_{pair}
NAB-Artificial	6	.545	.445	.613	+0.168
NAB-AWS	17	.400	.558	.549	-.009
NAB-AdExchange	5	.615	.590	.624	+0.034
NAB-Traffic	7	.615	.688	.779	+0.091
NAB-Tweets	10	.597	.656	.648	-.009
NASA-MSL	27	.543	.692	.684	-.008
NASA-SMAP	53	.726	.778	.760	-.017
Yahoo-A1	67	.614	.628	.657	+0.029
Yahoo-A2	100	.892	.876	.888	+0.013
Yahoo-A3	100	.614	.663	.669	+0.005
Yahoo-A4	100	.565	.415	.411	-.003
Equal-11	492	.612	.635	.662	+0.027

Yahoo-A1 (+0.0768) and Yahoo-A2 (+0.0584); the largest AUPRC loss is -0.0116 on NAB-Tweets. VUS-PR gains most on Yahoo-A1 (+0.0367) and Yahoo-A2 (+0.0360), whereas its largest loss is -0.0053 on NASA-SMAP. F1-max is more heterogeneous: IHP gains +0.1680 on NAB-Artificial and +0.0913 on NAB-Traffic but loses -0.0174 on NASA-SMAP. This heterogeneity is consistent with the wider F1-max interval and its arm-specific maximizing operating point.

B. External Paper-Reported Context

Table II separates the paper-reported ViT4TS values from the two local same-cache arms. IHP obtains equal-subdataset F1-max 0.662 versus the external ViT4TS value 0.612 and is numerically higher on 9/11 subdatasets. However, the local released-coordinate control already obtains 0.635; consequently, the external +0.050 difference is not the IHP effect. The attributable local difference is the paired +0.027 reported above.

The full paper-reported VLM4TS system reaches 0.659, but it adds an external language-model verification stage absent from both local arms. We therefore do not use it as a paired comparator. Yahoo-A4 further illustrates the environment boundary: IHP is 0.154 below the external ViT4TS entry but only 0.003 below the local control, while its same-cache AUPRC and VUS-PR increase by 0.0157

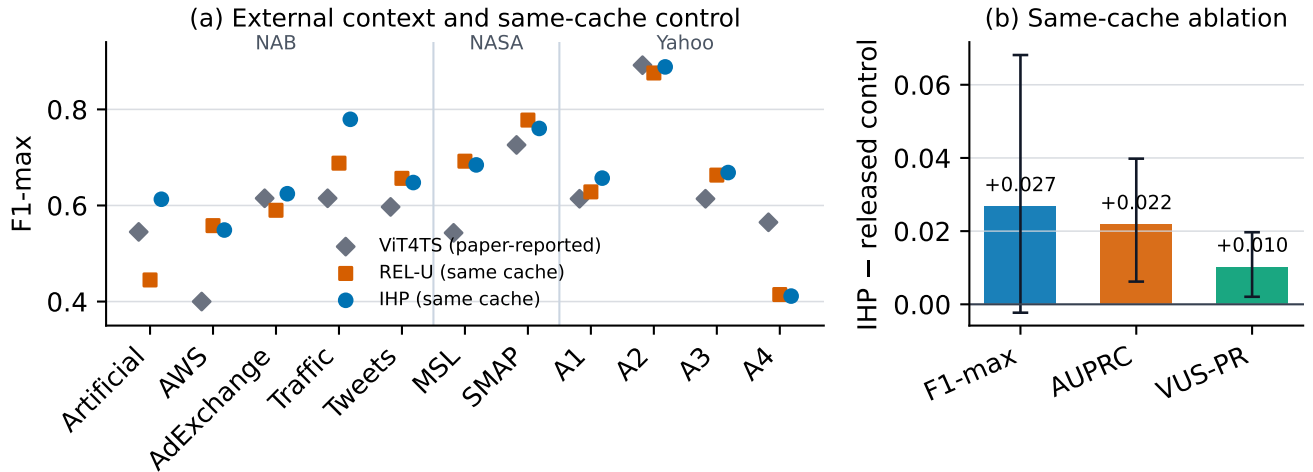


Fig. 2. Full evidence summary. (a) External paper-reported ViT4TS and the two local same-cache F1-max arms across eleven categorical subdatasets; the markers are intentionally unconnected. (b) Same-cache IHP-minus- REL-U differences with unadjusted hierarchical 95% intervals. The F1-max interval crosses zero; the AUPRC and VUS-PR intervals do not.

TABLE III

Label-free support-graph certificate on the 14×14 grid. “Rel. valid” counts shifted queries returning support; all 195 are displaced. “Row wraps” is the subset crossing a row boundary.

Scale	Output	Rel. valid	IHP valid	Row wraps	Holes
Medium	196	195	196	13	1
Large	196	195	196	13	1

and 0.0069. This discrepancy is compatible with threshold and environment differences and prevents a causal reading of the external rows.

C. Structural Certificate and Computational Boundary

Table III tests the mechanism without labels. At both pooled scales, every one of the 195 released queries that finds support is shifted by one flattened cell. Thirteen of those shifts wrap across row boundaries, and the terminal query is unsupported. Literal incidence covers all 196 output cells. The medium and large mask hashes are identical across all 492 cached transactions, so the certificate is not inferred from a single selected series.

IHP introduces zero trainable parameters, consumes no labels, and reuses the same CLIP token cache. All 492 paired manifests agree in cache key, token hash, cache-manifest hash, and encoder-call count. Replacing REL-U with IHP therefore adds zero token-cache bytes and preserves the persisted score footprint (11.82 MiB for all 492 series). It also adds no encoder pass or external-VLM call; only the fixed incidence lookup changes before the inherited harmonic reduction. Runtime and peak allocated VRAM were recorded jointly for three experimental arms, so we make no arm-isolated latency, VRAM, or host-RAM claim. The exact certificate and paired threshold-free improvements support the narrower conclusion that

the projection interface, rather than additional model capacity, changes the anomaly ranking.

VI. Limitations and Conclusion

This work audited whether a frozen visual anomaly screen returns multiscale evidence to the base-grid cells that generated it. The released ViT4TS projector shifts every valid pooled-scale query by one flattened cell and leaves the terminal query unsupported. IHP provides the minimal repair: it uses the supplied zero-based memberships literally and then applies the unchanged harmonic projector on the corrected support. In the 492-series same-cache comparison, IHP improves equal-subdataset F1-max, AUPRC, and VUS-PR by 0.0267, 0.0218, and 0.0102. Unadjusted intervals exclude zero for AUPRC and VUS-PR but not F1-max. The paper-reported ViT4TS value of 0.612 and IHP’s 0.662 provide secondary external context, not a paired method effect.

The evidence is bounded to the univariate ViT4TS screening stage, its released 14×14 grid and pooling masks, and three benchmark families. IHP was a prespecified component promoted after a composite candidate failed on the same evaluation, so its intervals are not selection-adjusted. Moreover, released full-series preprocessing and all-window median memory make the evaluated screen offline and transductive; IHP does not remove this deployment limitation. We do not evaluate downstream VLM verification, multivariate renderers, online drift, or other visual backbones. Future work should make coordinate contracts executable across other renderers and test them on an independent, streaming-compatible protocol. Within the present boundary, the exact 196/196 coverage certificate and paired ranking gains show that auditing an inverse-coordinate interface can improve a frozen screen

without training, labels, learned parameters, or another encoder pass.

References

- [1] Q. Liu and J. Paparrizos, “The elephant in the room: Towards a reliable time-series anomaly detection benchmark,” in *Advances in Neural Information Processing Systems*, vol. 37. Curran Associates, Inc., 2024, pp. 108 231–108 261.
- [2] Z. Dong, Y. Huang, S. Sun, W. Cui, and Z. Wang, “BLADE: Behavior-level anomaly detection using network traffic in web services,” in *2025 21st International Conference on Mobility, Sensing and Networking (MSN)*. IEEE, 2025, pp. 221–228.
- [3] Z. He, S. Alnegheimish, and M. Reimherr, “Harnessing vision-language models for time series anomaly detection,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 26, pp. 21 690–21 698, 2026.
- [4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. *Proceedings of Machine Learning Research*, vol. 139. PMLR, 2021, pp. 8748–8763. [Online]. Available: <https://proceedings.mlr.press/v139/radford21a.html>
- [5] J. Park and S. Kang, “PaAno: Patch-based representation learning for time-series anomaly detection,” in *The Fourteenth International Conference on Learning Representations*, 2026. [Online]. Available: <https://openreview.net/forum?id=NXThkM7Iym>
- [6] S. Xing, J. Niu, and T. Ren, “DoKnowAD: Calibrating normal representations with refined domain knowledge to enhance time series anomaly detection,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 32, pp. 27 126–27 134, 2026.
- [7] X. Sun, H. Zhou, and C. Li, “Multivariate time series anomaly detection with idempotent reconstruction,” in *Advances in Neural Information Processing Systems*, vol. 38. Curran Associates, Inc., 2025, pp. 160 900–160 958.
- [8] B. Li, Q. Shentu, Y. Shu, H. Zhang, M. Li, N. Jin, B. Yang, and C. Guo, “CrossAD: Time series anomaly detection with cross-scale associations and cross-window modeling,” in *Advances in Neural Information Processing Systems*, vol. 38. Curran Associates, Inc., 2025, pp. 137 959–137 985.
- [9] T. Yin, S. Fu, Z. Zhang, L. Huang, X. Zhang, Y. Yang, K. Yang, and M. Yan, “ScatterAD: Temporal-topological scattering mechanism for time series anomaly detection,” in *Advances in Neural Information Processing Systems*, vol. 38. Curran Associates, Inc., 2025, pp. 52 266–52 297.
- [10] Q. Shentu, B. Li, K. Zhao, Y. Shu, Z. Rao, L. Pan, B. Yang, and C. Guo, “Towards a general time series anomaly detector with adaptive bottlenecks and dual adversarial decoders,” in *The Thirteenth International Conference on Learning Representations*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/ca7998666c2e53cc1e882b7268414d8a-Abstract-Conference.html
- [11] X. Wu, X. Qiu, Z. Li, Y. Wang, J. Hu, C. Guo, H. Xiong, and B. Yang, “CATCH: Channel-aware multivariate time series anomaly detection via frequency patching,” in *The Thirteenth International Conference on Learning Representations*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/2b25c39788e5cf11d3541de433ebf4c0-Abstract-Conference.html
- [12] J. W. Park, K. Jung, D. Lee, H. Lee, D. Gwak, C. Park, J. Choo, and J. Cho, “Delving into large language models for effective time-series anomaly detection,” in *Advances in Neural Information Processing Systems*, vol. 38. Curran Associates, Inc., 2025, pp. 103 013–103 060.
- [13] X. Xu, H. Wang, Y. Liang, P. S. Yu, Y. Zhao, and K. Shu, “Can multimodal LLMs perform time series anomaly detection?” in *Proceedings of the ACM Web Conference 2026*, 2026, pp. 5392–5403.
- [14] Z. Wang, C. Pei, Y. Liu, H. Jiang, Q. Zhou, H. Si, H. Cui, J. Li, G. Xie, J. Li, and D. Pei, “ViTs: Teaching machines to see time series anomalies like human experts,” in *Proceedings of the ACM Web Conference 2026*, 2026, pp. 5286–5297.
- [15] Y. Yang, T. Lan, Y. Gao, Y. Lu, W. He, M. Wang, C. Liu, and C. Zhang, “VETime: Vision enhanced zero-shot time series anomaly detection,” 2026, arXiv preprint. [Online]. Available: <https://arxiv.org/abs/2602.16681>
- [16] Z. Gu, B. Zhu, G. Zhu, Y. Chen, M. Tang, and J. Wang, “UniVAD: A training-free unified model for few-shot visual anomaly detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 15 194–15 203.
- [17] J. Zhang, G. Wang, Y. Jin, and D. Huang, “Towards training-free anomaly detection with vision and language foundation models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 15 204–15 213.
- [18] C. Lendering, E. Akdag, and E. Bondarau, “SubspaceAD: Training-free few-shot anomaly detection via subspace modeling,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 28 557–28 566.
- [19] Y. Lee, H. Lim, S. Jang, and H. Yoon, “UniFormaly: Towards task-agnostic unified framework for visual anomaly detection,” *Pattern Recognition*, vol. 169, p. 111820, 2026.
- [20] H. Michaeli and D. Soudry, “Alias-free ViT: Fractional shift invariance via linear attention,” in *Advances in Neural Information Processing Systems*, vol. 38. Curran Associates, Inc., 2025, pp. 1473–1499.
- [21] R. A. Rojas-Gomez, T.-Y. Lim, M. N. Do, and R. A. Yeh, “Making vision transformers truly shift-equivariant,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 5568–5577.
- [22] J. Paparrizos, P. Boniol, T. Palpanas, R. S. Tsay, A. J. Elmore, and M. J. Franklin, “Volume under the surface: A new accuracy evaluation measure for time-series anomaly detection,” *Proceedings of the VLDB Endowment*, vol. 15, no. 11, pp. 2774–2787, 2022.
- [23] H. Tan, S. Tan, and J. Chen, “Unsupervised concept drift detection via generative distribution alignment,” in *2025 21st International Conference on Mobility, Sensing and Networking (MSN)*. IEEE, 2025, pp. 367–374.
- [24] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://openreview.net/forum?id=YicbFdNTTy>